# Weapon-Target Assignment Problem

CSE 462: Algorithm Engineering Sessional

Checkpoint 2 Presentation


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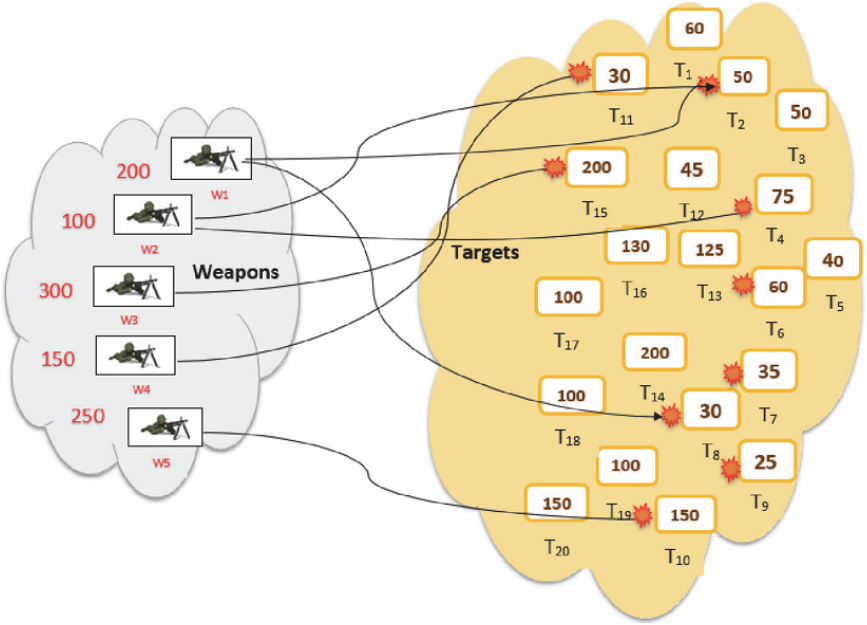
# Real-World Motivation: WTA in Action

**Air Defense (Iron Dome)**




**Ground Attack Scenario**

**WTA Mapping:**

- Interceptor Missiles = **Weapons**
- Incoming Rockets = **Targets**
- → Assignment = **WTA Solution**

**WTA Mapping:**

- W1–W5 (with costs) = **Weapons**
- T1–T20 (with values) = **Targets**
- → Arrows = **Assignments**

# Presentation Outline

## Part 1: Theory & Background

1. **Problem Definition**
   - Formal definition & formulation
   - Classifications
2. **NP-Hardness Proofs**
   - Reduction from SUBSET PRODUCT
   - Reduction to Multiple Knapsack
3. **Algorithm Survey**
   - Exact, approximation, heuristic, metaheuristic

## Part 2: Implementation & Results

4. **Selected Algorithms**
   - Maximum Marginal Return (MMR)
   - Genetic Algorithm (GA)
5. **Our Modifications**
   - MMR-IR: local search enhancement
   - Hybrid GA: 3 novel modifications
6. **Experiments & Results**
   - 180 instances, 9 plots, statistics

## What is the Weapon-Target Assignment Problem?

A fundamental combinatorial optimization problem in operations research

# Problem Definition

## Weapon-Target Assignment (WTA) Problem

The WTA problem involves the **strategic allocation of a set of weapons to a set of targets** to achieve a specific defensive or offensive objective.

## Core Challenge: Stochastic Nature of Combat

Every engagement is modeled as a **probabilistic event** where a weapon has a specific likelihood of destroying its assigned target.

## Computational Complexity

- The problem is mathematically proven to be **NP-complete**
- As the number of weapons and targets grows, complexity increases **exponentially**
- Finding an absolute optimal solution becomes extremely difficult for large instances

# Classification: Asset-based vs. Target-based

## Two Perspectives of WTA

The WTA problem can be viewed from two distinct perspectives depending on whether the objective prioritizes **destruction of enemy forces** or **protection of friendly resources**.

## Target-based Perspective

Values assigned to each **enemy target** based on threat they pose.

- **Goal:** Maximize total damage inflicted on incoming targets
- **Math: Minimize** expected survival value of targets
- (Equivalent to maximizing damage)

## Asset-based Perspective

Values assigned to **defended assets** (cities, bases) rather than targets.

- **Goal:** Maximize combined survival value of friendly assets
- **Requirement:** Map which targets attack which assets
- **Math: Maximize** expected value of surviving assets

# Mathematical Representation: Parameters and Variables

## Target-based Formulation – Parameters

- $W = \{1, \ldots, m\}$: The set of available **weapons**
- $T = \{1, \ldots, n\}$: The set of **targets** to be engaged
- $V_j$: The **strategic military value** or importance of target $j$
- $p_{ij}$: The "**probability of kill**" – chance that weapon $i$ will destroy target $j$

## Decision Variable

$$x_{ij} \in \{0, 1\}$$

**Binary variable:** $x_{ij} = 1$ if weapon $i$ is assigned to target $j$, 0 otherwise.
Each weapon is assigned to exactly one target.

# Mathematical Representation: Objective Functions

## Goal: Maximize damage (or equivalently, minimize survival)

While the goal is to maximize damage, the standard convention is to minimize remaining survival value.

## 1. Maximizing Total Expected Damage

$$\text{Max } Z = \sum_{j=1}^{n} V_j \left[ 1 - \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}} \right]$$

## 2. Minimizing Expected Survival Value

$$\text{Min } Z = \sum_{j=1}^{n} V_j \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$$

## Key Insight

$\prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$ = **Probability that target $j$ survives ALL assigned weapons**
The product captures the **combined effect** of multiple weapons attacking the same target.

# Mathematical Representation: Constraints

## Constraint 1: Each Weapon Assigned Once

$$\sum_{j=1}^{n} x_{ij} = 1, \quad \forall i \in W$$

**Meaning:** Every weapon must be assigned to exactly one target.

## Constraint 2: Integrality

$$x_{ij} \in \{0, 1\}$$

**Meaning:** Binary assignment – weapon $i$ is either assigned to target $j$ or not.

## Complete Formulation Summary

**Minimize** $\sum_j V_j \prod_i (1 - p_{ij})^{x_{ij}}$ **s.t.** $\sum_j x_{ij} = 1, \quad x_{ij} \in \{0, 1\}$

# Linear vs. Non-Linear Assignment

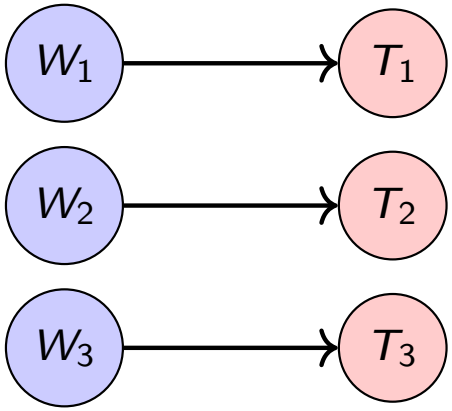
## Linear Assignment Problem (LAP)

- **Structure:** One weapon assigned to exactly one target (**One-to-One**)
- **Linearity:** Objective function is linear
- Assignments are **independent**
- **Polynomial time solvable** (Hungarian algorithm)

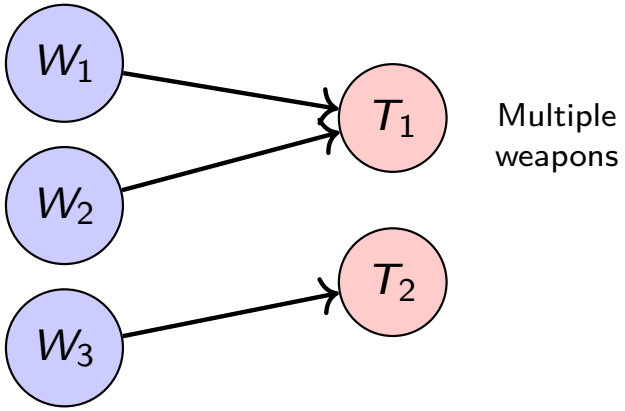
## Non-Linear Assignment (Non-LAP)

- **Structure:** Multiple weapons can be assigned to single target (**Many-to-One**)
- **Non-Linearity:** Due to "**diminishing returns**"
- Additional weapons = **overkill** effect
- **NP-complete!**

**LAP (One-to-One)**




**Non-LAP (Many-to-One)**



## Polynomial-Time Reductions

Proving WTA is NP-complete through formal reductions

# WTA Decision Problem

## Decision Version of WTA

**Given:**

- Survival probability $q_{i,j} = 1 - p_{i,j}$
- Threshold value $k$

**Question:** Does there exist an assignment $x_{i,j} \in \{0, 1\}$ such that:

$$D = \sum_{j=1}^{n} V_j \prod_{i=1}^{m} q_{i,j}^{x_{i,j}} \leq k$$

## Subject to Constraints

$$\sum_{j=1}^{n} x_{i,j} \leq W_i \quad \text{for } i = 1, \ldots, m$$

$$x_{i,j} \in \{0, 1\}$$

where $x_{i,j} = 1$ if weapon $i$ assigned to target $j$, and $x_{i,j} = 0$ otherwise.
$D$ represents **total expected damage** (weighted survival probability of all targets)

# Two Polynomial-Time Reductions

## Our Approach

We demonstrate two polynomial-time reductions:

## Reduction 1

**SUBSET PRODUCT $\leq_p$ WTA**

$\Rightarrow$ Proves **WTA is NP-hard**

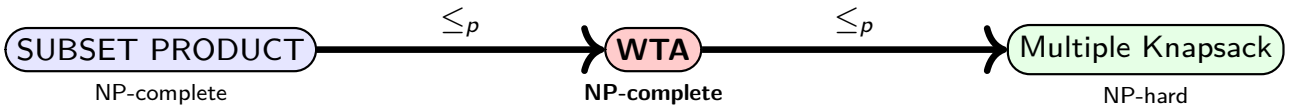
(Reduce FROM a known NP-complete problem TO WTA)

## Reduction 2

**WTA $\leq_p$ Multiple Knapsack**

$\Rightarrow$ Shows WTA reduces to another known NP-hard problem

(Shows connection between problems)



# Reduction 1: SUBSET PRODUCT $\leq_p$ WTA

## SUBSET PRODUCT Problem (NP-complete)

**Given:** Set $A = \{a_1, a_2, \ldots, a_m\}$ and target product $b$
**Find:** Subset $X \subseteq A$ where $\prod_{x \in X} a_x = b$

## WTA Instance Construction

Let $P = \prod_{i=1}^{m} a_i$. Construct WTA with:

- **Targets:** 2 targets with values $V_1 = V_2 = P$
- **Weapons:** $m + 2$ weapons: $W_1, W_2, \ldots, W_m, W_\alpha, W_\beta$
- **Survival probabilities:**
  - $q_{i,1} = q_{i,2} = \frac{1}{a_i}$ for $i = 1, \ldots, m$
  - $q_{\alpha,1} = q_{\alpha,2} = \frac{b}{2P}$
  - $q_{\beta,1} = q_{\beta,2} = \frac{1}{2b}$
- **Threshold:** $k = 1$ (we ask if $D \leq 1$)

## Time Complexity

Constructs WTA with 2 targets and $m + 2$ weapons in $O(m)$ time – **Polynomial!**

# Reduction 1: Forward Direction

**Forward ($\Rightarrow$): SUBSET PRODUCT has solution $\Rightarrow$ WTA has solution**

Suppose $X \subseteq \{1, \ldots, m\}$ with $\prod_{x \in X} a_x = b$. Let $Y = \{1, \ldots, m\} \setminus X$.
Then:

$$\prod_{i \in X} \frac{1}{a_i} = \frac{1}{b}, \quad \prod_{y \in Y} \frac{1}{a_y} = \frac{b}{P}$$

**Assignment:** $\{W_x : x \in X\} \cup \{W_\alpha\} \to T_1$ and $\{W_y : y \in Y\} \cup \{W_\beta\} \to T_2$
**Total damage:**

$$\begin{aligned} D &= V_1 \cdot \frac{b}{2P} \cdot \frac{1}{b} + V_2 \cdot \frac{1}{2b} \cdot \frac{b}{P} \\ &= P \cdot \frac{b}{2P} \cdot \frac{1}{b} + P \cdot \frac{1}{2b} \cdot \frac{b}{P} = \frac{1}{2} + \frac{1}{2} = 1 \leq k \quad \checkmark \end{aligned}$$

# Reduction 1: Backward Direction (Part 1)

## Backward ($\Leftarrow$): WTA has solution $\Rightarrow$ SUBSET PRODUCT has solution

Suppose WTA has solution with $D \leq 1$. Let $X', Y'$ be weapons assigned to $T_1, T_2$.

## Claim: $W_\alpha$ and $W_\beta$ cannot both target the same target

*Proof:* If both assigned to $T_1$ (symmetric for $T_2$):

$$D \geq P \cdot \frac{1}{2b} \cdot \frac{b}{2P} \cdot \frac{1}{\prod_{x \in X' \setminus \{\alpha,\beta\}} a_x} + \frac{P}{\prod_{y \in Y'} a_y} \geq \frac{1}{4} + 1 > 1 \quad \Rightarrow\Leftarrow$$

So $W_\alpha \to T_1$ and $W_\beta \to T_2$. Let $X = X' \setminus \{\alpha\}$, $Y = Y' \setminus \{\beta\}$.

## From the Assignment

With $W_\alpha \to T_1$ and $W_\beta \to T_2$:

$$D = P \cdot \frac{b}{2P} \prod_{x \in X} \frac{1}{a_x} + P \cdot \frac{1}{2b} \prod_{y \in Y} \frac{1}{a_y} \leq 1$$

# Reduction 1: Backward Direction (Part 2)

## Algebraic Derivation

Let $z = \prod_{x \in X} a_x$. Then:

$$\frac{b}{2z} + \frac{z}{2b} \leq 1$$

Multiply by $2bz > 0$:

$$b^2 + z^2 \leq 2bz \quad \Rightarrow \quad (b - z)^2 \leq 0$$

Since $(b - z)^2 \geq 0$, we must have $(b - z)^2 = 0$, so:

$$b = z = \prod_{x \in X} a_x$$

## Conclusion

$X$ is a valid solution to SUBSET PRODUCT! ✓
**Time Complexity:** $O(m)$ – Polynomial reduction complete.

# Reduction 2: WTA $\leq_p$ Multiple Knapsack

## Multiple Knapsack Construction

Given WTA with $m$ targets, $n$ weapons, probabilities $\{p_{i,j}\}$:

- **Knapsacks:** $m$ knapsacks (one per target) with Capacity $C_j = \infty$
- **Items:** $w_i$ items for weapon type $i$ (one per copy)
- **Profit:** $v_{ij} = -\log(1 - p_{i,j}) = -\log(q_{i,j})$
- **Question:** Does assignment with value $\geq V$ exist?

## Objective Transformation (Key Insight)

For single target $j$, minimizing $\prod q_{i,j}^{x_{i,j}}$ equals minimizing:

$$\log\left(\prod_{i=1}^{n} q_{i,j}^{x_{i,j}}\right) = \sum_{i=1}^{n} x_{i,j} \log(q_{i,j})$$

Equivalent to **maximizing**:

$$-\sum_{i=1}^{n} x_{i,j} \log(q_{i,j}) = \sum_{i=1}^{n} x_{i,j} \cdot v_{ij}$$

This is **exactly the Multiple Knapsack objective!**

# Reduction 2: Constraint Mapping & Conclusion

## Constraint Mapping

- **WTA constraint:** $\sum_{j=1}^{m} x_{i,j} \leq W_i$ (weapon type $i$ used at most $W_i$ times)
- **Maps to:** $\sum_{j=1}^{m} y_{ij} \leq 1$ (each item in at most one knapsack) when $W_i = 1$
- For general $W_i$: Create $W_i$ copies of item $i$

## Equivalence

Assignment $x_{i,j}$ optimal for WTA $\Leftrightarrow$ $y_{ij} = x_{i,j}$ optimal for Multiple Knapsack
**Logarithmic transformation preserves ordering of solutions!**
**Time:** $O(nm)$ to compute all $v_{ij} = -\log(1 - p_{i,j})$

## Final Conclusion

**WTA is in NP** (solution verifiable in polynomial time) + **WTA is NP-hard**
$\Rightarrow$ **The Weapon-Target Assignment Problem is NP-complete!**

## Comprehensive Algorithm Categories for WTA

Five major categories of solution approaches

### Exact

- Optimal solutions
- Exponential time
- Small instances

### Approximation

- Provable bounds
- Polynomial time
- Large instances

### Randomized

- Probabilistic
- Quality $\propto$ samples
- Baseline approach

### Heuristic

Fast, problem-specific, near-optimal

### Metaheuristic

General-purpose, exploration/exploitation balance

# Category 1: Exact Exponential Algorithms

## Guarantee Optimal Solutions with Proven Bounds

These algorithms guarantee finding the optimal solution but have exponential time complexity.

| Algorithm | Description | Complexity | Solution Quality | Ref |
|---|---|---|---|---|
| **Branch-and-Bound (B&B)** | Uses LP/IP relaxations with branching on weapon-target assignments. Employs maximum marginal return for branching strategy and network flow for lower bounds. | Worst-case: $O(2^{mn})$ nodes | Optimal; handles up to $80 \times 80$ instances | [2] |
| **Branch-Price-and-Cut (BPC)** | Reformulates WTA for column generation with efficient pricing problem solver. Uses clique cuts and power cone optimization for tight relaxations. | Polynomial per iteration; $O(nm \cdot K)$ | Optimal; solves 10000+ in seconds | [5] |

# Category 2: Approximation Algorithms

## Provable Bounds with Polynomial Time Complexity

These algorithms provide theoretical guarantees on solution quality while running in polynomial time.

| Algorithm | Description | Complexity | Solution Quality | Ref |
|---|---|---|---|---|
| **Piecewise Linear Approx.** | Approximates nonlinear survival probability using piecewise linear convex functions with breakpoints. | $O(mn \cdot B)$ | Provable $\delta$-error bounds | [6] |
| **LP Relaxation + Rounding** | Relaxes integer constraints to LP, solves continuous problem, then rounds to feasible integer solution. | $O(\text{poly}(m, n))$ | No guarantee; may violate constraints | [2] |
| **Lagrangian Relaxation** | Relaxes weapon constraints via Lagrangian multipliers, decomposes into independent target subproblems. | $O(T \cdot nm)$ | Provides lower bounds; typically 5-15% gap | [7] |

# Categories 3 & 4: Randomized and Heuristic Algorithms

## Randomized Algorithms

| Algorithm | Description |
|---|---|
| **Monte Carlo** | Generates random assignments, evaluates each, selects best. $O(S \cdot mn)$ |
| **Constructive** | Probabilistic weapon-target selection based on marginal returns with randomness. |

## Heuristic Algorithms

| Algorithm | Description |
|---|---|
| **MMR** | Greedy: max marginal return per step. $O(W^2 T)$, near-optimal [1] |
| **VLSN** | Very Large-Scale Neighborhood search via improvement graphs [2] |

## General-Purpose Optimization Frameworks

Metaheuristics are adaptable frameworks that balance exploration and exploitation.

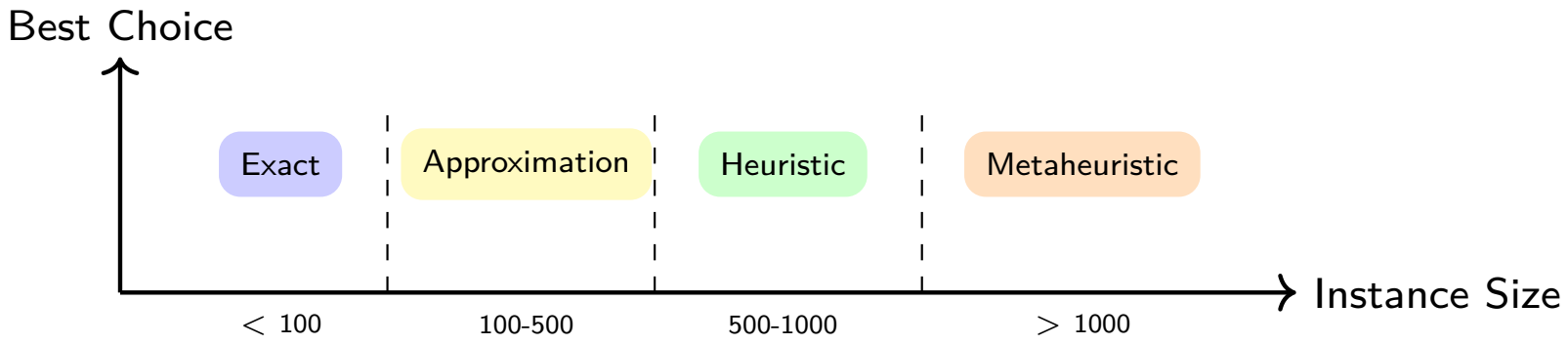
| Algorithm | Description | Complexity | Solution Quality | Ref |
|---|---|---|---|---|
| **Genetic Algorithm (GA)** | Population-based evolution with crossover and mutation operators. Uses tournament selection and fitness evaluation. | $O(G \cdot P \cdot mn)$ $G$=gens, $P$=pop | Good for medium-large; 5-10% gap | [3] |
| **Ant Colony (ACO)** | Pheromone-based probabilistic construction. Transition probability weighted by heuristic info. | $O(I \cdot A \cdot mn)$ $I$=iters, $A$=ants | Competitive; avoids premature convergence | [9] |
| **Simulated Annealing** | Temperature-based acceptance of worse solutions for escaping local optima. Gradual cooling schedule. | $O(I \cdot mn)$ | Good quality; sensitive to cooling schedule | [8] |

# Complexity Comparison Summary

## When to Use Each Category?

The choice depends on instance size, time constraints, and solution quality requirements.

| Category | Typical Complexity | Instance Size Limit | Use Case |
|---|---|---|---|
| Exact | $O(2^{mn})$ - Exponential | $< 1000$ variables | When optimality required |
| Approximation | $O(\text{poly}(m, n))$ | $> 10000$ | Need theoretical bounds |
| Randomized | $O(S \cdot mn)$ | Any (quality $\propto S$) | Baselines, diversity |
| Heuristic | $O(mn)$ - $O(m^2 n^2)$ | $> 1000$ | Real-time systems |
| Metaheuristic | $O(I \cdot P \cdot mn)$ | $> 10000$ | Large-scale optimization |



## Maximum Marginal Return (MMR) and Genetic Algorithm (GA)

Detailed algorithms with step-by-step examples

### **MMR** – Greedy Heuristic

- Fast: $O(W^2 T)$ complexity
- Deterministic output
- Near-optimal (<5% gap)
- **Ideal for real-time systems**

### **GA** – Metaheuristic

- Flexible: $O(G \cdot P \cdot W \cdot T)$
- Population-based exploration
- Avoids local optima
- **Ideal for large instances**

# Why MMR + GA?

## Maximum Marginal Return (MMR)

**Why selected:**

- $O(W^2 T)$ – fastest complexity
- Simple to implement and verify
- Often near-optimal (<5% gap)
- **Ideal for real-time missile defense**

**Key Formula:**

$$\Delta_{ij} = V_j \cdot q_j \cdot (1 - q_{ij})$$

(Marginal reduction in survival)

## Genetic Algorithm (GA)

**Why selected:**

- Population-based global search
- Natural selection drives optimization
- Well-studied for combinatorial problems
- **Effective for large instances (>100)**
- Easy to incorporate domain knowledge

**Key Idea:**
Evolve population of solutions through selection, crossover, and mutation

## Complementary Approaches

**MMR** provides fast, deterministic baseline; **GA** performs global stochastic search across diverse solution space. Together, they represent greedy vs. evolutionary paradigms.

# Algorithm 1: Maximum Marginal Return (MMR)

## Core Idea

**Greedy approach:** At each step, assign the unallocated weapon and target pair that yields the **maximum decrease** in survival value.

## Key Concepts

- **Tracking Value:** We track remaining survival value for each target.
- **Decrease Calculation:** decrease $= V[t] \times p[w][t]$
- **Value Update:** Once assigned, the target's value is permanently reduced:

$$V[t] \leftarrow V[t] - \text{decrease}$$

- **Iteration:** Repeat until all weapons are allocated.

## Example: Diminishing Returns

If T1 has $V = 10$, W1 ($p = 0.7$) yields decrease $= 7.0$. New $V = 10 - 7 = 3.0$. A second weapon W3 ($p = 0.8$) on T1 now yields decrease $= 3.0 \times 0.8 = 2.4$.

# MMR Algorithm: Pseudocode

**Algorithm 1** Maximum Marginal Return (MMR) Algorithm

```
solution.Allocations ← ∅
solution.Value ← MaxValue
allocatedWeaponCount ← 0
while allocatedWeaponCount ≤ noOfWeapons do
    maxDecrease ← MinValue
    k ← 1
    while k ≤ unallocatedWeapons.Count do
        i ← 1
        while i ≤ noOfTargets do
            decrease ← targetValues[i] × killProbabilities[i][k]
            if decrease > maxDecrease then
                maxDecrease ← decrease
                allocatedTarget ← i
                allocatedWeapon ← k
            end if
            i ← i + 1
        end while
        k ← k + 1
    end while
    unallocatedWeapons.Remove(allocatedWeapon)
    solution.Allocations[allocatedWeapon] ← allocatedTarget
    targetValues[allocatedTarget] ← targetValues[allocatedTarget] − maxDecrease
    allocatedWeaponCount ← allocatedWeaponCount + 1
end while
```

# MMR Example: Problem Setup (Naval Air Defense)

**Incoming Missiles (Targets):**

- T1: Aircraft Carrier ($V = 10$)
- T2: Destroyer ($V = 6$)
- T3: Supply Ship ($V = 3$)

**Defensive Weapons:**

- W1, W2, W3, W4 (4 interceptors)

**Goal:** Maximize expected damage prevented

**Kill Probability Matrix $P_{ij}$:**

| | T1 | T2 | T3 |
|---|---|---|---|
| W1 | 0.7 | 0.8 | 0.9 |
| W2 | 0.6 | 0.7 | 0.8 |
| W3 | 0.8 | 0.6 | 0.7 |
| W4 | 0.5 | 0.9 | 0.6 |

**Objective Function:**

$$\text{Max} \sum_j V_j \times P_{kill,j}$$

# MMR Iteration 1: Calculate Initial Marginal Returns

**Algorithm Step:** Calculate marginal return for all weapon-target pairs

**Formula:** $MR_{ij} = V_j \times P_{ij}$ (since no weapons assigned yet)

| | T1 ($V = 10$) | T2 ($V = 6$) | T3 ($V = 3$) |
|---|---|---|---|
| W1 | $10 \times 0.7 = 7.0$ | $6 \times 0.8 = 4.8$ | $3 \times 0.9 = 2.7$ |
| W2 | $10 \times 0.6 = 6.0$ | $6 \times 0.7 = 4.2$ | $3 \times 0.8 = 2.4$ |
| W3 | $10 \times 0.8 = \mathbf{8.0}$ | $6 \times 0.6 = 3.6$ | $3 \times 0.7 = 2.1$ |
| W4 | $10 \times 0.5 = 5.0$ | $6 \times 0.9 = 5.4$ | $3 \times 0.6 = 1.8$ |

## MMR Choice

**Highest Marginal Return = 8.0** $\Rightarrow$ Assign **W3** $\rightarrow$ **T1**

**Current State:** T1: $P_{kill} = 0.8$, Expected $= 8.0$ **Remaining:** W1, W2, W4

# MMR Iteration 2: Recalculate with Diminishing Returns

**Algorithm Step:** Update marginal returns accounting for W3 already on T1

**For T1 (already has W3, $P_{kill} = 0.8$):**

- W1: $P'_{kill} = 1 - (0.3 \times 0.2) = 0.94$, $MR = 10(0.94 - 0.8) = 1.4$
- W2: $P'_{kill} = 1 - (0.4 \times 0.2) = 0.92$, $MR = 10(0.92 - 0.8) = 1.2$
- W4: $P'_{kill} = 1 - (0.5 \times 0.2) = 0.90$, $MR = 10(0.90 - 0.8) = 1.0$

| | T1 (has W3) | T2 (empty) | T3 (empty) |
|---|---|---|---|
| W1 | 1.4 | 4.8 | 2.7 |
| W2 | 1.2 | 4.2 | 2.4 |
| W4 | 1.0 | **5.4** | 1.8 |

## MMR Choice

**Highest Marginal Return = 5.4** $\Rightarrow$ Assign **W4** $\rightarrow$ **T2**

**Current State:** T1: 8.0, T2: $6 \times 0.9 = 5.4$ **Total: 13.4**

# MMR Iterations 3 & 4: Complete Assignment

**Iteration 3:** Marginal returns for W1, W2

| | T1 (has W3) | T2 (has W4) | T3 (empty) |
|---|---|---|---|
| W1 | 1.4 | 0.48 | **2.7** |
| W2 | 1.2 | 0.42 | 2.4 |

$\Rightarrow$ Assign **W1** $\rightarrow$ **T3** (MR $= 2.7$)

**Iteration 4:** Only W2 remains

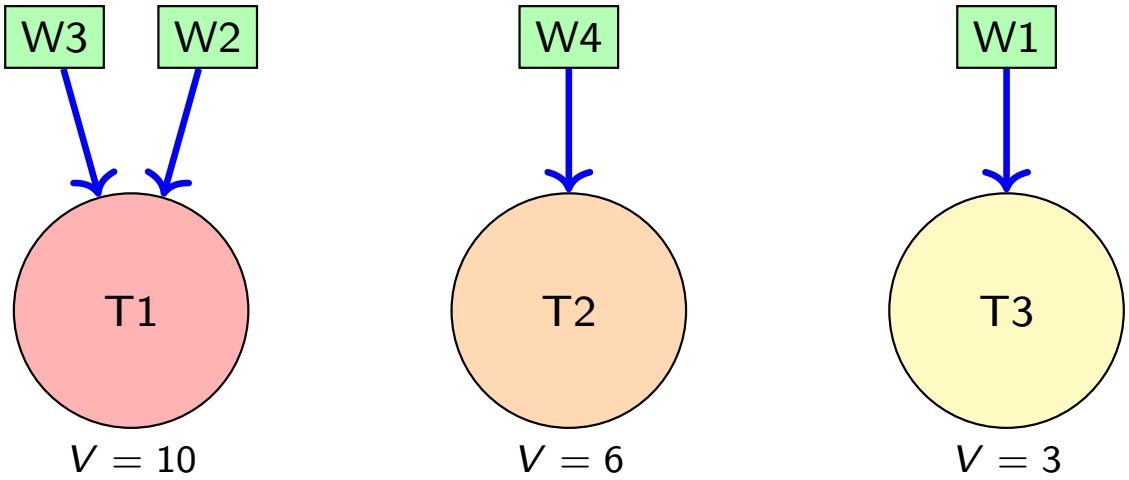
| | T1 (has W3) | T2 (has W4) | T3 (has W1) |
|---|---|---|---|
| W2 | **1.2** | 0.42 | 0.24 |

$\Rightarrow$ Assign **W2** $\rightarrow$ **T1** (MR $= 1.2$)

## Note: Diminishing Returns

T3 with W1 has $P_{kill} = 0.9$, so adding W2: $P'_{kill} = 1 - (0.2 \times 0.1) = 0.98$, $MR = 3(0.98 - 0.9) = 0.24$

# MMR: Final Solution




## Final Assignment & Expected Value

- **T1 (Aircraft Carrier):** W2, W3 $\rightarrow P_{kill} = 1 - (0.4 \times 0.2) = 0.92$, EV $= 10 \times 0.92 = 9.2$
- **T2 (Destroyer):** W4 $\rightarrow P_{kill} = 0.9$, EV $= 6 \times 0.9 = 5.4$
- **T3 (Supply Ship):** W1 $\rightarrow P_{kill} = 0.9$, EV $= 3 \times 0.9 = 2.7$

## Total Expected Value (Damage Prevented)

$9.2 + 5.4 + 2.7 = \mathbf{17.3}$ (out of max $10 + 6 + 3 = 19$)

# MMR: Algorithm Analysis

## Advantages

- **Fast:** $O(W^2 \cdot T)$ time complexity
- **Simple:** Easy to implement
- **Real-time capable:** Suitable for time-critical defense systems
- **Good results:** Often near-optimal
- **Adaptive:** Considers current state when making decisions

## Disadvantages

- **No optimality guarantee:** May miss globally optimal solution
- **Greedy nature:** Early decisions can't be reversed
- **Local optimization:** Focuses on immediate best choice

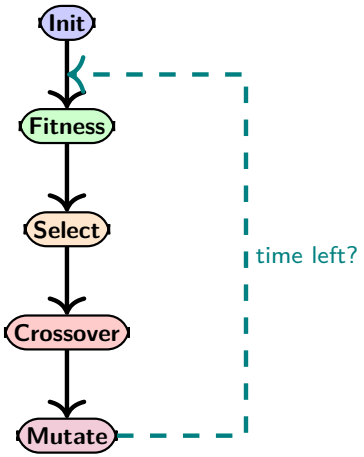
## When to Use MMR

- Speed is critical
- Near-optimal is acceptable
- Real-time defense systems
- Large-scale problems

# Algorithm 2: Genetic Algorithm (GA)

## Core Idea

**Bio-inspired approach:** Evolve a population of candidate solutions through natural selection – fitter individuals survive and reproduce.




## GA for WTA

- **Chromosome Encoding:** target assigned to weapon $i$
- **Population Size:** $pop_size = \max(W, T)$
- **Fitness:** $-Z$ (negative survival value; higher = better)
- **Selection:** Tournament ($k = 3$)
- **Crossover:** Uniform ($p_{crossover} = 0.8$)
- **Mutation:** Random ($p_{mutation} = 1/W = 0.33$)

## Key Advantage

Explores **many solutions simultaneously** – can escape local optima that trap greedy methods

# GA: Encoding for WTA

## Chromosome Representation

Each individual (chromosome) is a vector of length $W$ (number of weapons):

$$\text{chromosome}[i] = j \quad \Rightarrow \quad \text{weapon } i \text{ assigned to target } j$$

| | $W_0$ | $W_1$ | $W_2$ |
|---|---|---|---|
| **Chromosome:** | **2** | **0** | **1** |

**Meaning:** $W_0 \to T_2$, $W_1 \to T_0$, $W_2 \to T_1$

## Properties

- Fixed-length: always $W$ genes
- Each gene $\in \{0, 1, \ldots, T-1\}$
- Multiple weapons can target same target (many-to-one)
- Every weapon assigned to exactly one

## Fitness Function (Survival Value $Z$)

$$Z = \sum_{j=0}^{T-1} V_j \prod_{i:\text{chrom}[i]=j} (1 - p_{ij})$$

Lower $Z$ = lower survival = **better** solution

# GA Algorithm: Pseudocode

**Algorithm 2** Genetic Algorithm

```
startTime ← Now
endTime ← startTime + allowedSearchTime
solution.Allocations ← ∅,   solution.Value ← MaxValue
if noOfTargets > noOfWeapons then
    noOfIndividuals ← noOfTargets
else
    noOfIndividuals ← noOfWeapons
end if
population ← GenerateInitialPopulation(noOfIndividuals)
while Now < endTime do
    individualNo ← 1
    while individualNo ≤ noOfIndividuals do
        solFromIndv ← population[individualNo]
        solValue ← CalculateSolutionValue(solFromIndv)
        if solValue < solution.Value then
            solution ← solFromIndv
        end if
        individualNo ← individualNo + 1
    end while
    parents ← SelectParents(population)
    population ← CrossOver(parents)
    population ← Mutate(population)
end while
return solution
```

# GA Example: Problem Setup (3 Weapons, 3 Targets)

**Targets (with strategic values):**

- $T_0$: Value $V_0 = 10$
- $T_1$: Value $V_1 = 20$
- $T_2$: Value $V_2 = 30$

**Weapons:** $W_0, W_1, W_2$

**GA Parameters:**

- Population size $= \max(3, 3) = 3$
- Tournament $k = 3$, Crossover $= 0.8$
- Mutation rate $= 1/W = 1/3$

**Kill Probability Matrix $p_{ij}$:**

| | $T_0$ | $T_1$ | $T_2$ |
|---|---|---|---|
| $W_0$ | 0.3 | 0.2 | 0.1 |
| $W_1$ | 0.1 | 0.4 | 0.2 |
| $W_2$ | 0.2 | 0.3 | 0.5 |

**Fitness (Survival Value $Z$):**

$$Z = \sum_j V_j \prod_{i:x_i=j} (1 - p_{ij})$$

Lower $Z$ = better solution

# GA: Generation 0 – Initialization & Fitness

**Three random individuals are generated:**

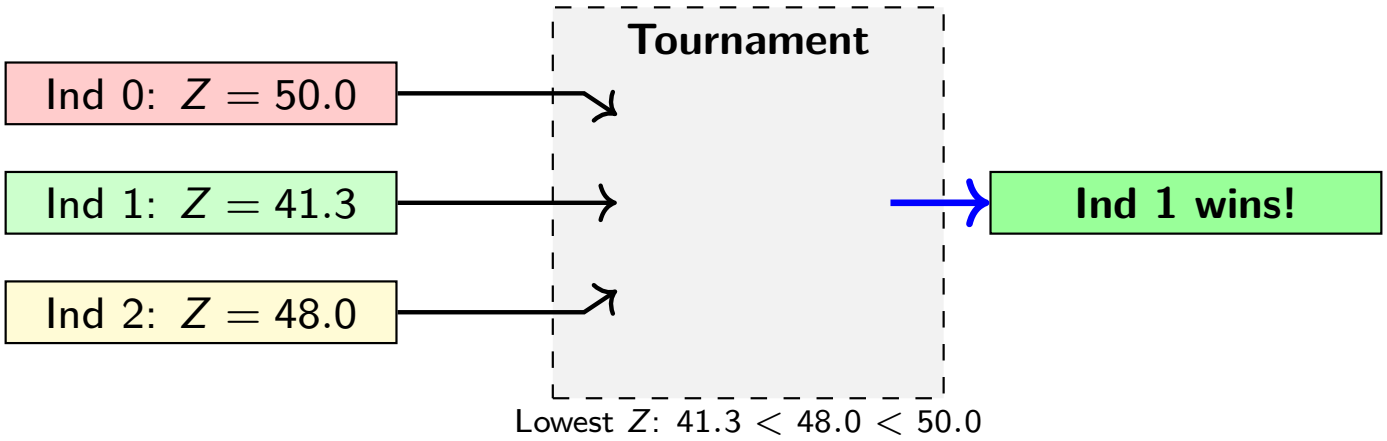
| | | | | | |
|---|---|---|---|---|---|
| **Ind 0:** | 2 | 0 | 1 | $W_0 \to T_2,\ W_1 \to T_0,\ W_2 \to T_1$ | $Z = 50.0$ |
| **Ind 1:** | 0 | 0 | 2 | $W_0 \to T_0,\ W_1 \to T_0,\ W_2 \to T_2$ | $Z = \mathbf{41.3}$ |
| **Ind 2:** | 1 | 2 | 0 | $W_0 \to T_1,\ W_1 \to T_2,\ W_2 \to T_0$ | $Z = 48.0$ |

## Detailed Fitness Calculation (Ind 1: $[0, 0, 2]$)

- $T_0$ attacked by $W_0, W_1$: $V_0 \times (1 - 0.3)(1 - 0.1) = 10 \times 0.7 \times 0.9 = 6.3$
- $T_1$ not attacked: $V_1 = 20.0$
- $T_2$ attacked by $W_2$: $V_2 \times (1 - 0.5) = 30 \times 0.5 = 15.0$
- $Z = 6.3 + 20.0 + 15.0 = \mathbf{41.3}$ (Best – lowest survival)

## Tournament Selection ($k = 3$)

Randomly pick $k$ individuals from the population; select the one with **lowest** $Z$ (best fitness).




## Why Tournament Selection?

- **Selection pressure** adjustable via $k$: larger $k$ = stronger pressure
- Preserves **diversity**: weaker individuals still have a chance
- $O(k)$ per selection – very efficient

## In This Example

Both parents happen to be the same individual (Ind 1: $[0, 0, 2]$), so crossover produces an identical child. **Mutation** then introduces variation.

## Uniform Crossover

$P_1 = [0, 0, 2]$, $P_2 = [0, 0, 2]$

Child (before mutation): $[0, 0, 2]$

Identical parents $\Rightarrow$ identical child. Crossover preserves the best solution's genes.

## Mutation (rate = 1/3)

Each gene mutates with probability 1/3:

- Gene 0: rand $= 0.71 > 0.33 \Rightarrow$ **Keep** 0
- Gene 1: rand $= 0.12 < 0.33 \Rightarrow$ **Mutate!** $0 \rightarrow 2$
- Gene 2: rand $= 0.55 > 0.33 \Rightarrow$ **Keep** 2

| | | | | |
|---|---|---|---|---|
| **Before:** | 0 | 0 | 2 | $[0, 0, 2]$: $W_0 \rightarrow T_0$, $W_1 \rightarrow T_0$, $W_2 \rightarrow T_2$ |
| | | ↑ mutate! | | |
| **After:** | 0 | **2** | 2 | $[0, \mathbf{2}, 2]$: $W_0 \rightarrow T_0$, $W_1 \rightarrow T_2$, $W_2 \rightarrow T_2$ |

# GA: Generation 1 – Offspring Evaluation & Replacement

## Offspring Fitness: Child $C_1 = [0, 2, 2]$

- $T_0$ attacked by $W_0$: $V_0 \times (1 - 0.3) = 10 \times 0.7 = 7.0$
- $T_1$ not attacked: $V_1 = 20.0$
- $T_2$ attacked by $W_1, W_2$: $V_2 \times (1 - 0.2)(1 - 0.5) = 30 \times 0.8 \times 0.5 = 12.0$
- $Z = 7.0 + 20.0 + 12.0 = \mathbf{39.0}$ (Better than any parent!)

## Replacement: Child Replaces Worst Individual

| | Before | | After |
|---|---|---|---|
| Ind 0 | $[2, 0, 1]$, $Z = 50.0$ | $\Rightarrow$ | $[0, 2, 2]$, $Z = \mathbf{39.0}$ |
| Ind 1 | $[0, 0, 2]$, $Z = 41.3$ | $\Rightarrow$ | $[0, 0, 2]$, $Z = 41.3$ |
| Ind 2 | $[1, 2, 0]$, $Z = 48.0$ | $\Rightarrow$ | $[1, 2, 0]$, $Z = 48.0$ |

New best: $Z = 39.0$ (improved from 41.3)

## Comparison with MMR on Same Problem

MMR greedy gives $[0, 1, 2]$ with $Z = 10(0.7) + 20(0.6) + 30(0.5) = 7 + 12 + 15 = \mathbf{34.0}$

After 1 generation, GA (39.0) is still behind MMR (34.0) – motivating **MMR-seeded initialization** in our Hybrid GA!

# GA: Parameters Summary

## GA Configuration (Hasan & Barua)

All parameters follow the original paper implementation.

| Parameter | Value | Rationale |
|---|---|---|
| Population size | $\max(W, T)$ | Scales with problem size |
| Selection | Tournament ($k = 3$) | Good selection pressure |
| Crossover rate | 0.8 | High recombination |
| Crossover type | Uniform | No positional bias |
| Mutation rate | $1/W$ | $\sim$1 gene per individual |
| Termination | Time-based | Fair comparison with MMR |

## Time Budgets for Fair Comparison

| Problem Size | Time Budget |
|---|---|
| Small ($W \leq 30$) | 5 seconds |
| Medium ($W \leq 120$) | 20 seconds |
| Large ($W > 120$) | 60 seconds |

## Improving MMR and GA with Novel Enhancements

3 modifications for each algorithm – all designed by our team

### MMR → MMR-IR

1. Tie-breaking by target value
2. 1-opt local search
3. 2-opt swap search

**Goal:** Escape greedy local optima

### GA → Hybrid GA

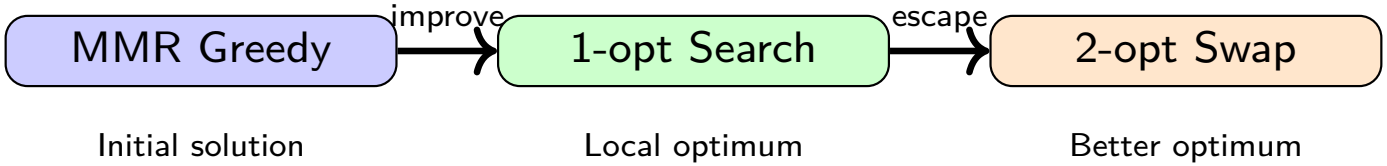
1. MMR-seeded initialization
2. Elitism + stagnation detection
3. Threat-proportional mutation

**Goal:** Faster convergence + better exploration

# Modification 1: MMR-IR (Iterative Refinement)

## Overview: Three Enhancements to MMR

The greedy MMR solution is a good starting point but may be suboptimal because early decisions cannot be reversed. Our modifications add **post-processing local search** to improve the greedy solution.

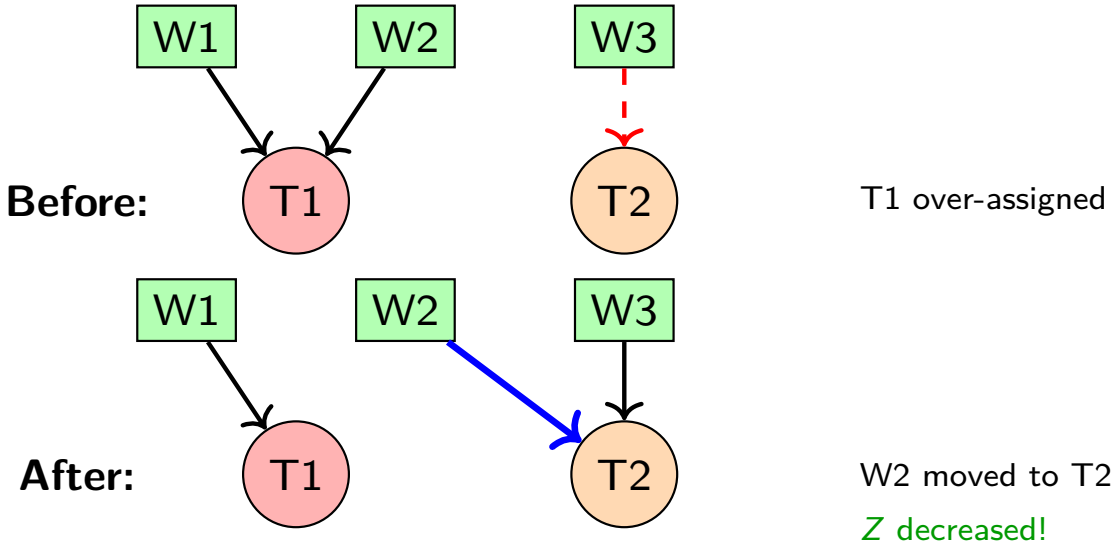



## Enhancement 1: Tie-Breaking by Target Value

When two weapon-target pairs have **equal marginal return**, prefer the pair targeting the **higher-value target**. This prioritizes coverage of critical targets.

## Enhancement 2: 1-opt Incremental Local Search

After MMR greedy completes, try **reassigning each weapon to every other target**. Accept if the objective improves. Repeat until no improvement found.

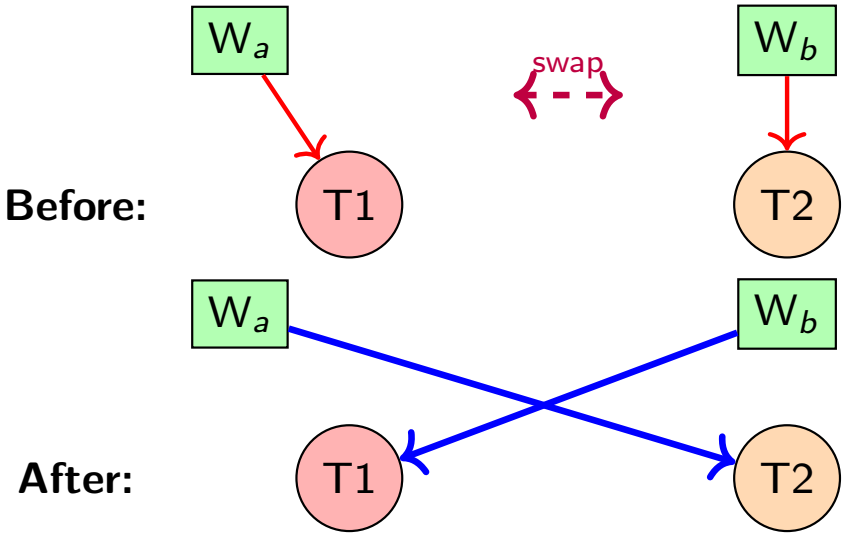



## Incremental Evaluation

Uses $O(1)$ delta computation per trial (survival array maintained via multiplication/division) instead of $O(WT)$ full recomputation. Max 20 passes until convergence.

## Enhancement 3: 2-opt Swap Search

After 1-opt converges, try **swapping the targets of two weapons simultaneously**. This can escape 1-opt local optima that single reassignments cannot.




## Why 2-opt Helps

- 1-opt is stuck: moving $W_a$ alone worsens T1, moving $W_b$ alone worsens T2
- But swapping **both simultaneously** can improve the total objective
- Max 3 passes to limit runtime; incremental $O(1)$ delta per trial

# Modification 2: Hybrid GA – Overview

## Three Weaknesses of Standard GA on WTA

1. **Random initialization** – in high-dimensional space ($W \times T$ variables), random solutions are far from optimal
2. **No elitism** – the best solution found can be **lost** when replaced by offspring
3. **Blind mutation** – random target reassignment ignores which targets **need** more weapons

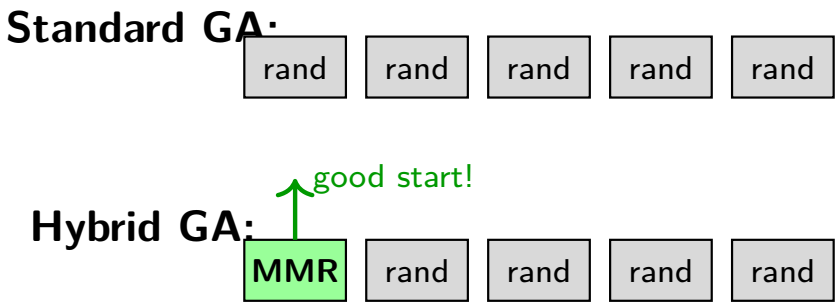
## Our Three Fixes

| **Weakness** | **Our Fix** | **Type** |
|---|---|---|
| Random initialization | MMR-seeded init | Warm start |
| No elitism / stagnation | Elitism + detection | Preservation |
| Blind mutation | Threat-proportional | **WTA-specific** |

# Hybrid GA Mod 1: MMR-Seeded Initialization

## Idea: Start from a Good Solution

Instead of initializing **all** individuals randomly, seed `population[0]` with the MMR greedy solution. The rest remain random for diversity.



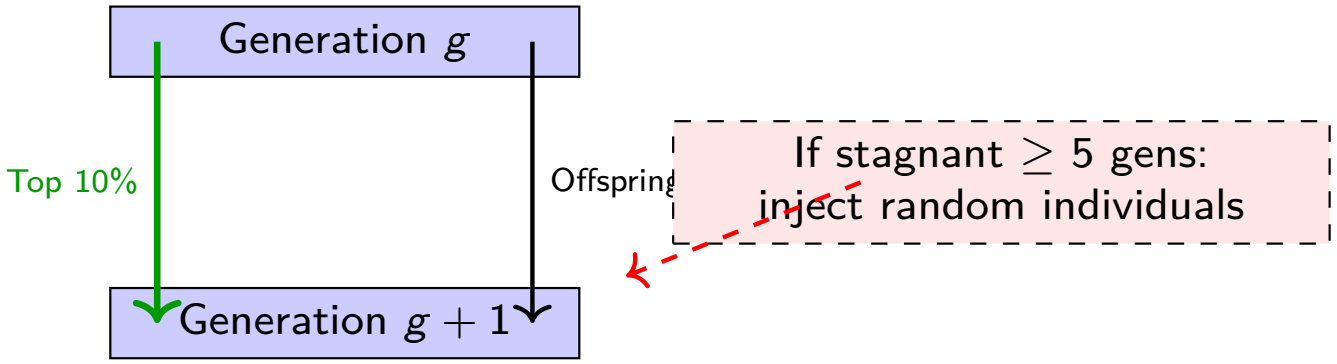

## Why This Works

- MMR gives a **near-optimal** starting point in $O(W^2 T)$ time
- GA then **refines** this solution via crossover and mutation
- Random individuals provide **genetic diversity** to avoid local optima
- For large instances ($W = 375$), this saves hundreds of wasted generations

# Hybrid GA Mod 2: Elitism + Stagnation Detection

## Idea: Never Lose the Best, Re-diversify When Stuck

- **Elitism:** Preserve top 10% of population each generation (directly copied to next generation without modification)
- **Stagnation detection:** If best fitness unchanged for 5 consecutive generations, inject fresh random individuals to re-diversify




## Without Elitism (Standard GA)

A good solution in generation $g$ can be completely lost if all offspring are worse. This is especially problematic with small time budgets.

# Hybrid GA Mod 3: Threat-Proportional Mutation

## Idea: Domain-Aware Mutation (Novel, WTA-Specific)

Instead of mutating to a **uniformly random** target, bias mutation toward targets with **high remaining threat**:

$$\text{remaining_threat}[j] = V_j \times \prod_{i:x_i=j} (1 - p_{ij})$$

**Mutation target selection:** $P(\text{mutate to } j) \propto \text{remaining_threat}[j]$

## Standard Mutation

- Pick random target uniformly
- May reassign to already well-covered target (wasted mutation)
- No domain knowledge used

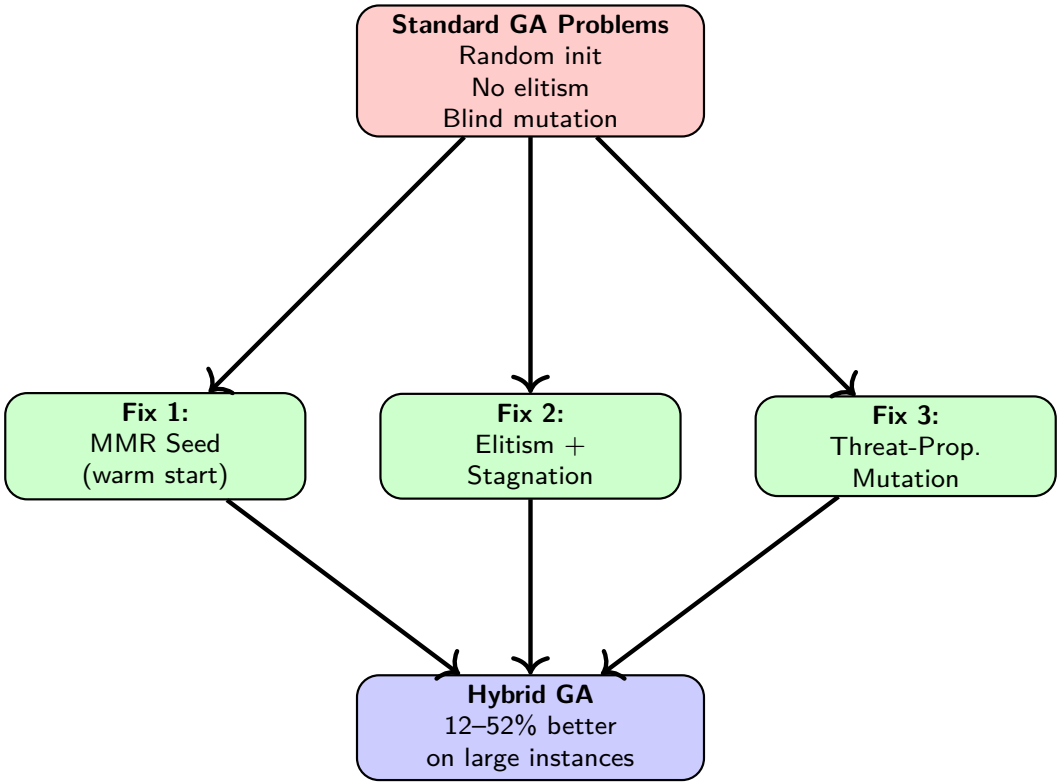
## Threat-Proportional Mutation

- High-value, poorly-covered targets get more mutations
- Focuses search where improvement is **most needed**
- **Novel:** This is our original WTA-specific contribution

## Example

If T1 ($V = 100$, survival=0.6) and T2 ($V = 20$, survival=0.1):
$\text{threat}_1 = 100 \times 0.6 = 60$, $\text{threat}_2 = 20 \times 0.1 = 2$
$\Rightarrow$ T1 is 30$\times$ more likely to receive a mutated weapon than T2

# Hybrid GA: Modifications Summary




## Key Insight

Each modification targets a **specific weakness**. Together, they transform a struggling standard GA into a competitive solver that matches or beats MMR on all problem sizes.

## Dataset Design, Setup, and Results

180 instances, 4 algorithms, 9 categories

# Dataset Design: Custom Synthetic Benchmark

## Why Design Our Own Dataset?

We created a synthetic dataset to systematically test **scaling behavior** across two dimensions: **problem size** (small/medium/large) and **resource availability** (balanced/scarce/rich).

| **Tier** | **Scenario** | **Weapons ($W$)** | **Targets ($T$)** | **Instances** |
|---|---|---|---|---|
| Small | Balanced ($W = T$) | 20 | 20 | 20 |
| Small | Scarce ($W < T$) | 10 | 20 | 20 |
| Small | Rich ($W > T$) | 30 | 20 | 20 |
| Medium | Balanced | 100 | 100 | 20 |
| Medium | Scarce | 50 | 100 | 20 |
| Medium | Rich | 150 | 100 | 20 |
| Large | Balanced | 250 | 250 | 20 |
| Large | Scarce | 125 | 250 | 20 |
| Large | Rich | 375 | 250 | 20 |

**Total: 180 instances** (9 categories $\times$ 20 instances each)

# Dataset Parameters and Scenarios

## Instance Parameters

- **Target values:** $V_j \sim U(10, 100)$ integers
- **Kill probabilities:** $p_{ij} \sim U(0.05, 0.90)$
- **RNG seed:** 42 (reproducible)
- **Format:** JSON files

## Statistical Testing

- **Wilcoxon signed-rank test** ($\alpha = 0.05$) for pairwise algorithm comparison
- 20 instances per category ensures statistical validity

## Scenario Rationale

**Balanced** ($W = T$): Standard benchmark; each weapon covers one target on average.

**Scarce** ($W < T$): Hardest scenario; not enough weapons for all targets. Unengaged targets have 100% survival.

**Rich** ($W > T$): Easiest scenario; surplus weapons allow thorough coverage. Lower objective values expected.

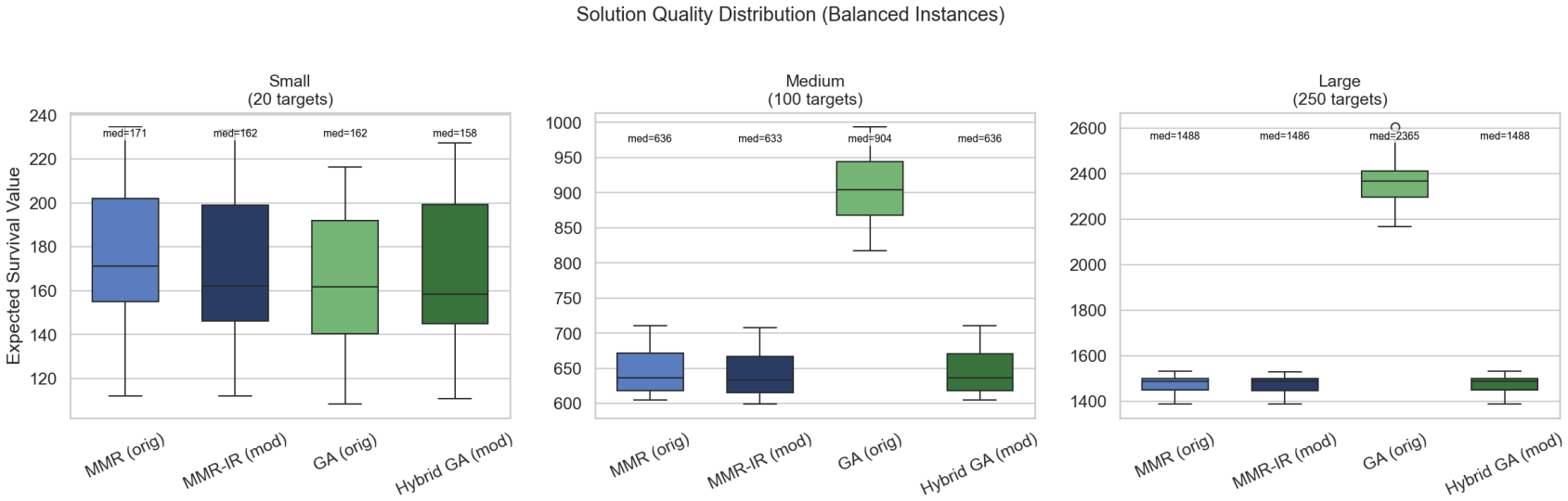
# Algorithms Compared

| Algorithm | Description | Type |
|---|---|---|
| **MMR (orig)** | Maximum Marginal Return greedy heuristic. Faithful implementation from Hasan & Barua. | Heuristic |
| **MMR-IR (mod)** | MMR + tie-breaking + 1-opt local search + 2-opt swap search. | Enhanced Heuristic |
| **GA (orig)** | Standard Genetic Algorithm. Faithful implementation from Hasan & Barua. | Metaheuristic |
| **Hybrid GA (mod)** | GA + MMR seed + elitism + stagnation detection + threat-proportional mutation. | Enhanced Meta. |

## Fair Comparison

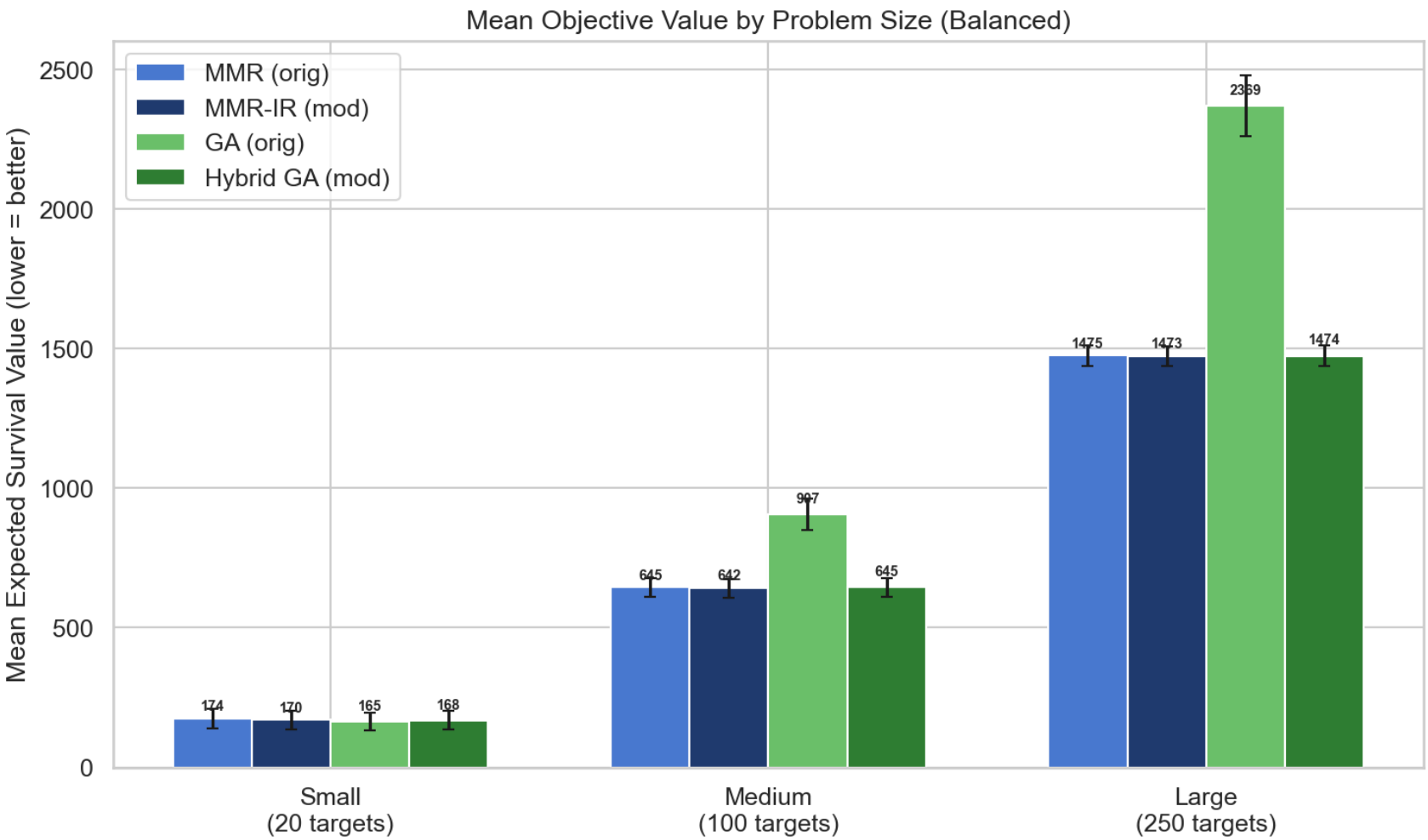
All algorithms run on the **same instances** with **same time budgets**. GA variants use identical time limits. Results measured as **expected survival value** $Z$ (lower = better).

# Results: Solution Quality – Box Plots



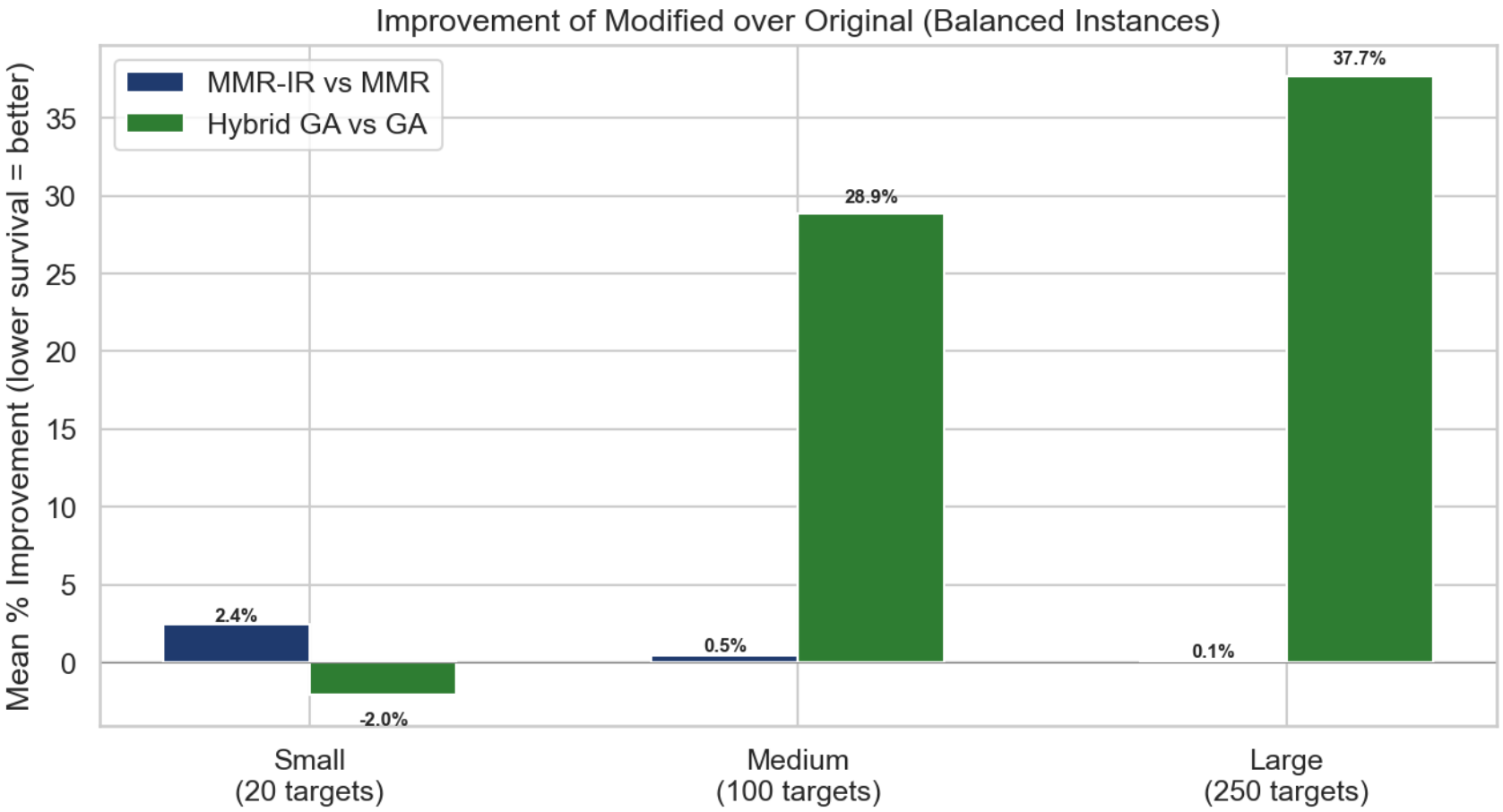

Box plots show distribution of objective values (lower = better) across balanced instances per tier.

# Results: Mean Objective Value by Tier



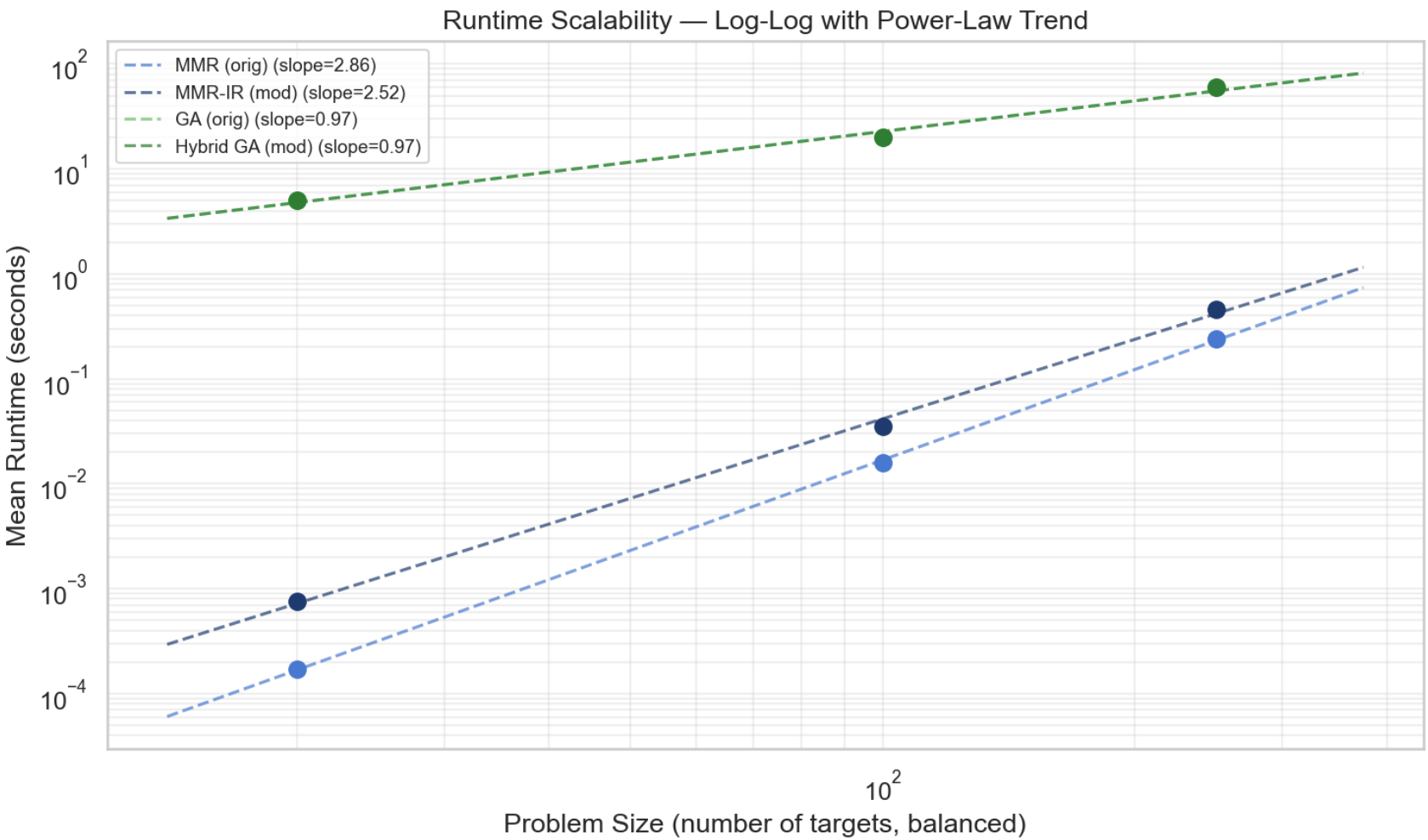

GA (orig) falls dramatically behind on medium and large tiers due to random initialization + no elitism.

# Results: Improvement of Modifications



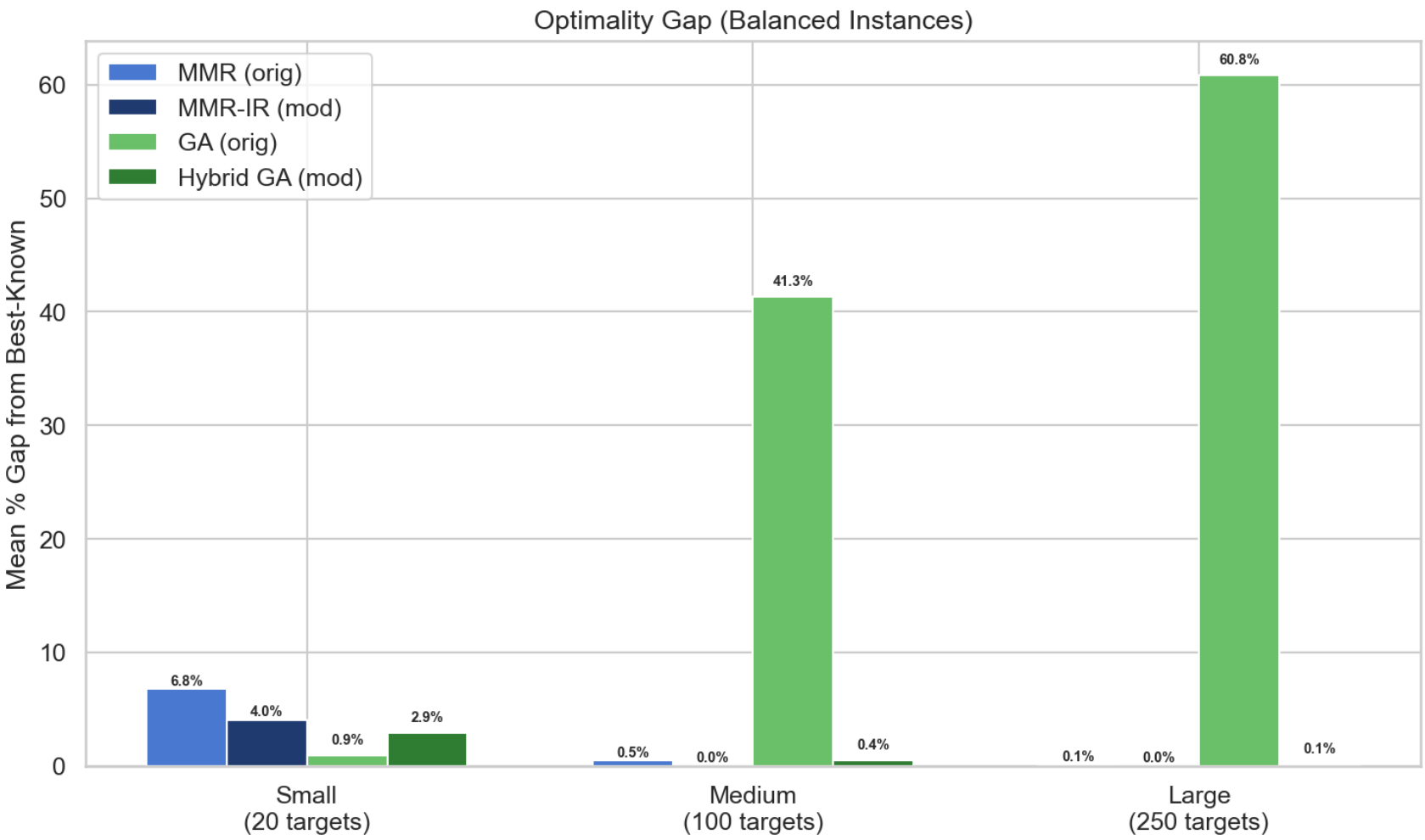

Percentage improvement of modified algorithms over their original counterparts. GA improvement grows with problem size.

# Results: Runtime Scalability



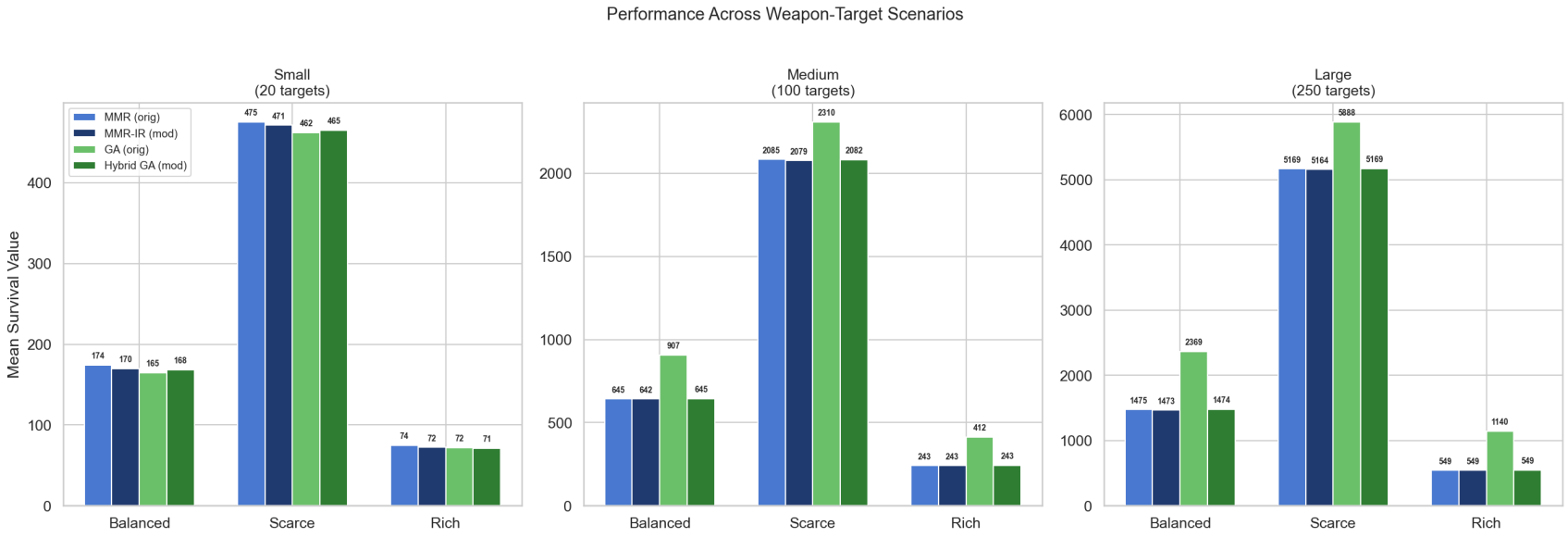

Log-log scale. MMR variants are orders of magnitude faster. GA variants use their full time budget.

# Results: Optimality Gap



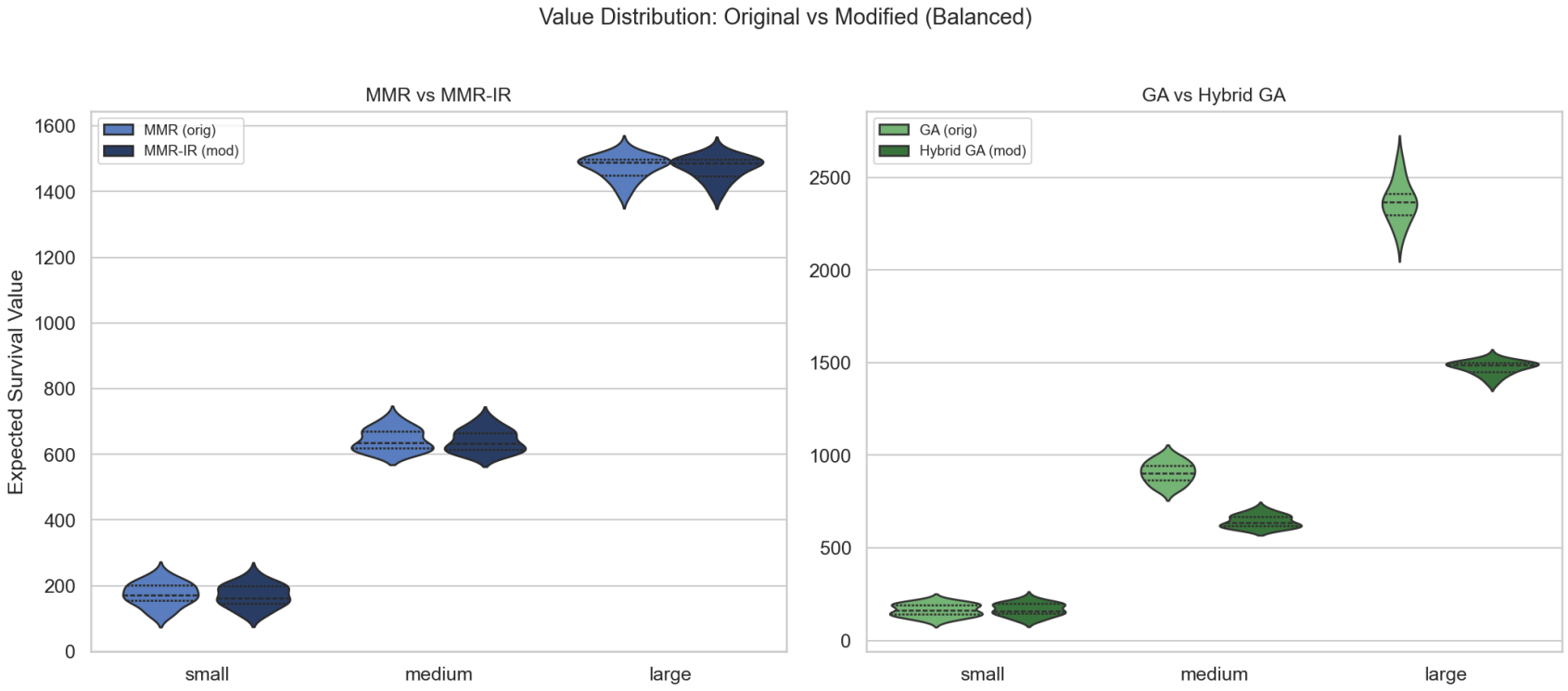

Gap from best-known solution. GA (orig) has the largest gap, especially on large instances.

# Results: Scenario Comparison



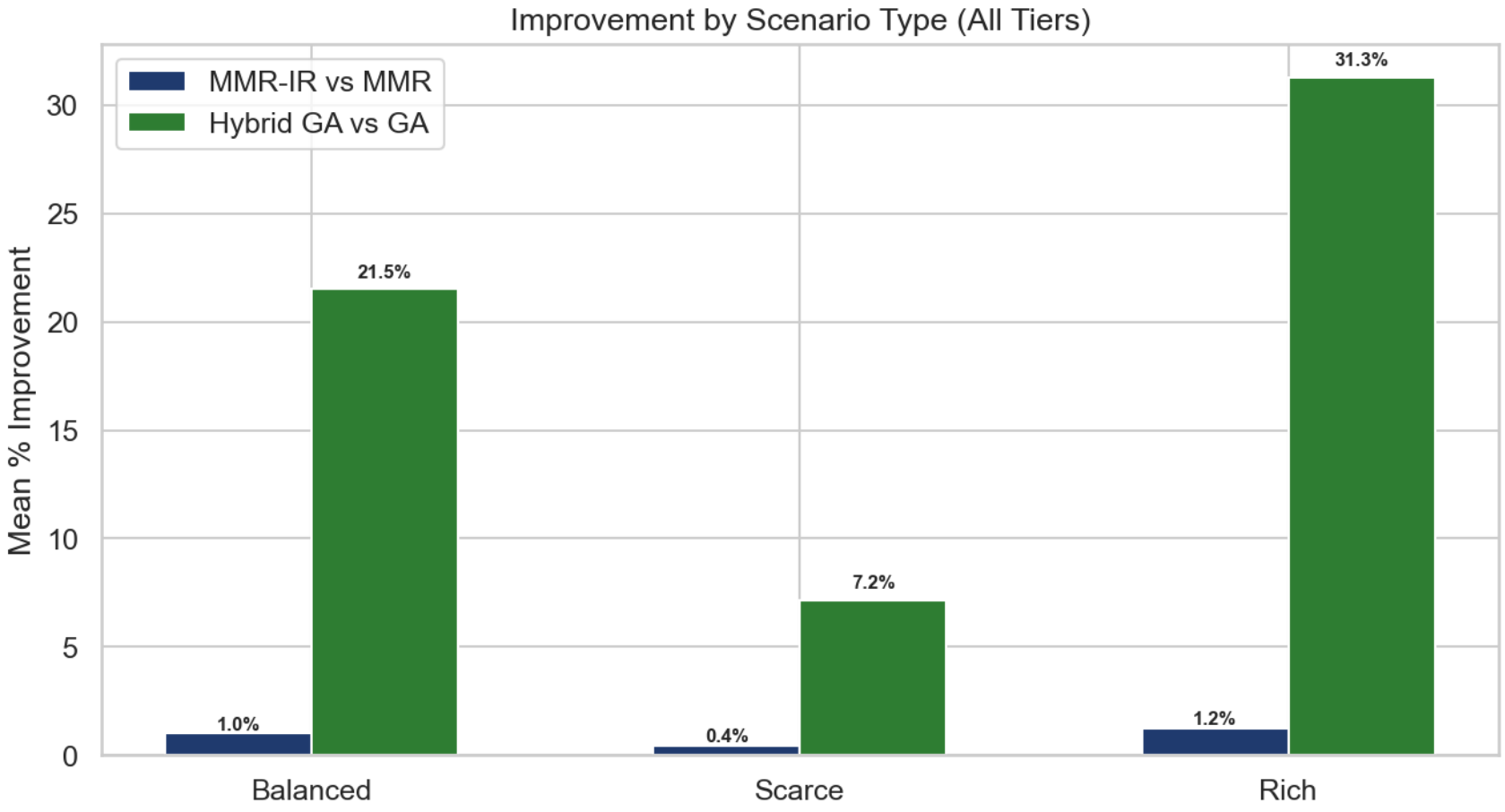

Performance across balanced, scarce, and rich scenarios. Scarce has highest objective (hardest), rich has lowest (easiest).

# Results: Value Distributions



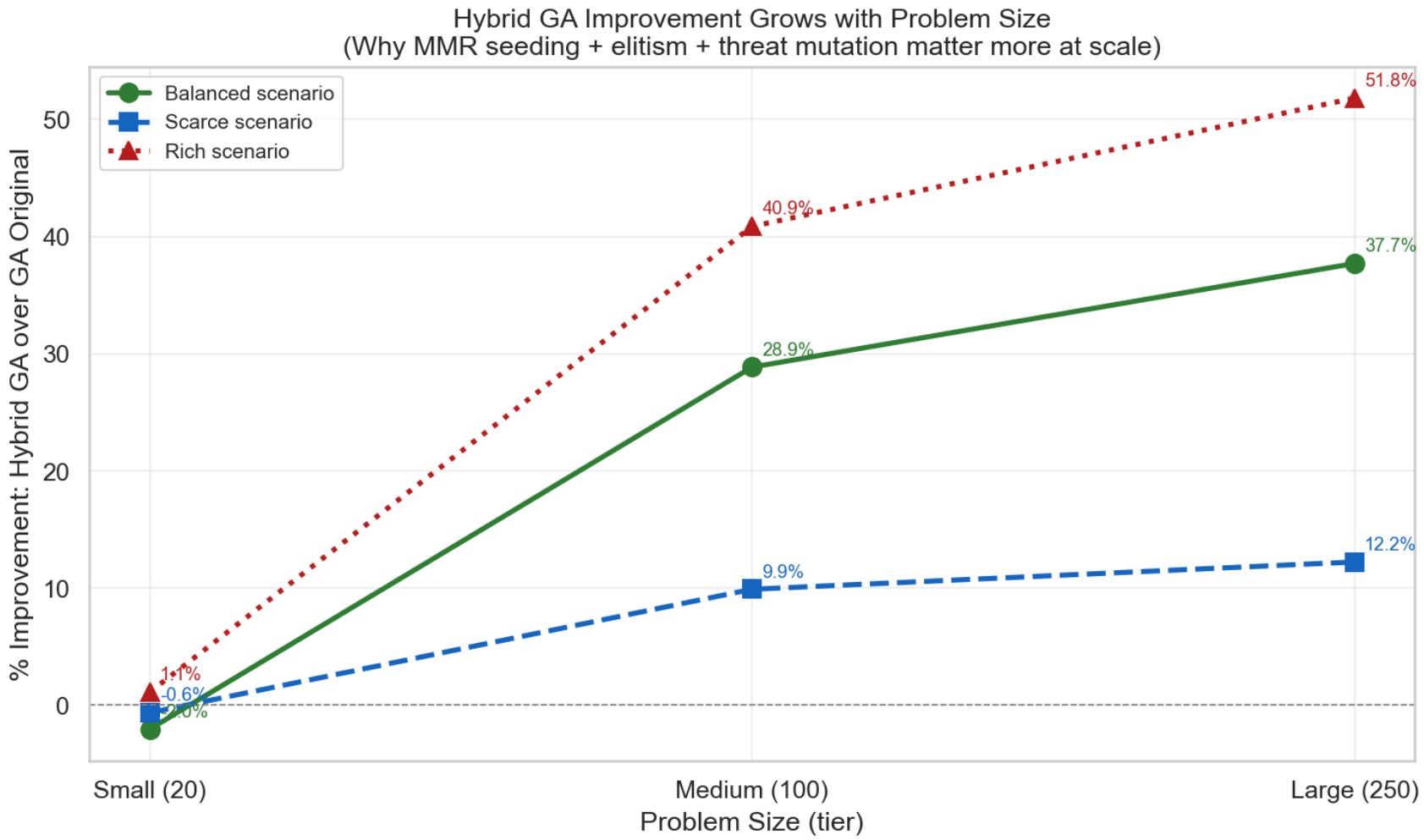

Violin plots showing the distribution of objective values for original vs. modified algorithms.

# Results: Improvement by Scenario




Improvement percentage broken down by scenario type (balanced, scarce, rich).

# Results: GA Improvement vs. Problem Size




**Key finding:** Hybrid GA improvement over standard GA grows dramatically with problem size – from ∼-2% (small) to ∼52% (large).

# Key Findings

## MMR-IR vs MMR (Original)

- **Consistent improvement:** 0.1–3.1% across all tiers and scenarios
- 1-opt and 2-opt local search reliably escape greedy suboptimality
- Negligible additional runtime (local search is $O(1)$ per trial)

## Hybrid GA vs GA (Original) – The Main Story

- **Small:** -2–1% improvement (modest)
- **Medium:** 10–41% improvement (significant)
- **Large: 12–52% improvement** (dramatic)
- Improvement **grows with problem size** – exactly where it matters most

## Statistical Significance

Wilcoxon signed-rank test ($\alpha = 0.05$): All pairwise comparisons between original and modified algorithms are **statistically significant** across all tiers.

# Why GA (Original) Fails at Scale

## Root Cause Analysis

Three compounding factors explain why standard GA underperforms on medium/large instances:

1. **Large population, limited time:**
   - Population size $= \max(W, T)$. For large_rich: $\max(375, 250) = 375$ individuals
   - With 60s time budget: only $\sim$160 generations completed
   - 160 generations are insufficient to evolve 375-dimensional random solutions
2. **Random initialization:**
   - All individuals start as random assignments – far from any good solution
   - In 375-dimensional space, random solutions have extremely high objective values
3. **No elitism:**
   - The best solution found in early generations can be **lost**
   - Replaced by offspring that may be worse

## Our Hybrid GA Fixes All Three

MMR seed (warm start) + Elitism (preservation) + Threat-proportional mutation (guided search)

# Results Summary Table

| Category | MMR (orig) | MMR-IR | GA (orig) | Hybrid GA |
|---|---|---|---|---|
| small_balanced | 174 | 170 | 165 | 168 |
| small_scarce | 475 | 471 | 462 | 465 |
| small_rich | 74.5 | 72.2 | 72.1 | 71.2 |
| medium_balanced | 645 | 642 | 907 | 645 |
| medium_scarce | 2085 | 2079 | 2310 | 2082 |
| medium_rich | 243 | 243 | 412 | 243 |
| large_balanced | 1475 | 1473 | 2369 | 1474 |
| large_scarce | 5169 | 5164 | 5888 | 5169 |
| large_rich | 549 | 549 | 1140 | 549 |

Mean objective values (lower = better). **Bold observation:** GA (orig) diverges dramatically on medium/large tiers. Hybrid GA brings it back to MMR-level performance. MMR-IR consistently improves over MMR by small but statistically significant margins.

# Reproducibility: GitHub Repository

## Single Master Script

All experiments are reproducible with a single command:

```
python run_all.py
```

Generates dataset → runs all 4 algorithms on 180 instances → produces 9 plots.
**Runtime:** ∼3.5 hours total.

## Repository Structure

```
CSE_462-Algorithm-Engineering-Sessional/
  codes/
    run_all.py           # Master script
    dataset_generator.py # 180 instances
    experiment_runner.py # Run all algorithms
    analysis.py          # 9 plots + stats
    mmr_original.py      # MMR (Hasan & Barua)
    mmr_modified.py      # MMR-IR (ours)
    ga_original.py       # GA (Hasan & Barua)
    ga_modified.py       # Hybrid GA (ours)
  datasets/              # JSON instances
  results/figures/       # 9 PNG plots
```

# GitHub Link

**GitHub Repository:**

https://github.com/Arimotokika/CSE_462-Algorithm-Engineering-Sessional

## Quick Test (5 instances, ∼1 minute)

```
python experiment_runner.py -categories small_balanced -instances 1-5
python analysis.py
```

# References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# Thank You!

Questions?